Retail Probability of Default Model: Development and Validation Report

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Model LC-PD-01, version 1.0. Self-review in the format of an independent validation; not independent.

Document control

Item	Detail
Model	LC-PD-01: lifetime probability of default, unsecured 36-month consumer instalment loans
Model type	Champion: WoE logistic scorecard. Challenger: gradient-boosted trees (LightGBM)
Intended uses	Application decisioning (cut-off), risk-based pricing input, expected loss, portfolio stress
Data	Lending Club accepted loans 2007–2018Q4 (public), 36-month loans issued to Dec 2015
Development sample	Issued Oct 2012 – Dec 2013: 115,603 loans, 12.4% default rate
Out-of-time samples	2014 (162,570 loans) and 2015 (282,787 loans)
Code	`credit-risk/` repository, package `lcrisk`; `lcrisk reproduce` rebuilds every number and figure here from the raw file
Review status	The author built and reviewed the model. This report follows the structure of an independent validation, but it is not independent, and a real deployment would need one.

1. Executive summary

The scorecard ranks risk consistently in and out of time: its Gini is 0.310 on the hold-out test sample, 0.316 on 2014 loans and 0.330 on 2015 loans, with no decay. Ten features carry it, all with the expected sign, all significant, with variance inflation factors below 2. On the development window it is well calibrated (Hosmer–Lemeshow p = 0.65; no master-scale grade rejected).

Out of time, calibration fails. Default rates rose inside every Lending Club grade after 2013, and the model under-predicts the 2015 vintage by two percentage points (mean PD 12.9%, observed 14.9%); the Jeffreys test rejects the PD of seven of eight grades. The population stability index never left green (score PSI ≤ 0.026 in every quarter), so **standard population monitoring would not have caught this.** The applicants looked the same on paper and defaulted more often. Recalibrating the intercept on 2014 outcomes, as an annual review would have done, closes only a third of the 2015 gap, because conditions kept deteriorating.

The LightGBM challenger adds about four Gini points in every sample (0.352 / 0.354 / 0.363). At a 20% decline rate on the 2015 book that is 628 more defaults avoided, roughly $4 million of loss on $3.6 billion lent. Its top decline reason agrees with the scorecard's only 29% of the time.

Opinion: approve the scorecard as champion, with conditions. (1) Recalibrate before use and on monitoring triggers, not on a fixed annual cycle. (2) Add an early-delinquency backtest to monitoring, because the 36-month outcome arrives too late to steer by. (3) Keep LightGBM as a challenger and benchmark until a governed reason-code framework exists for it. The findings log is in Section 10.

Gini (95% bootstrap interval)	Test (in time)	Out of time 2014	Out of time 2015
Scorecard (champion)	0.310 (0.292–0.326)	0.316 (0.308–0.324)	0.330 (0.325–0.335)
LightGBM (challenger)	0.352 (0.338–0.369)	0.354 (0.345–0.362)	0.363 (0.359–0.368)
Lending Club sub-grade (benchmark)	0.277 (0.259–0.293)	0.319 (0.312–0.326)	0.357 (0.353–0.361)

2. Purpose, scope and use

The model estimates the probability that a new 36-month unsecured instalment loan to an individual borrower is charged off at any point in its life (lifetime PD). It is built from information available at application only: the credit-bureau record, stated income and employment, the loan request and the debt-to-income ratio.

In scope: individual applications for 36-month loans in the product and credit-policy range of the development data (FICO 660 and above, Lending Club's minimum).

Out of scope: 60-month loans (different default timing; excluded from development), joint applications (239 in the whole 2007–2015 population, too few to model), behavioural scoring of loans already on book, and IFRS 9 / CECL provisioning. That would need a term structure of PD and forward-looking adjustments. Section 7 gives an empirical 12-month PD for the Basel-style calculation only.

3. Data

3.1 Source and lineage

The data are Lending Club's public loan-level files, 2007 to 2018Q4, as compiled on Kaggle (`wordsforthewise/lending-club`) and fetched from a CC0 mirror. The file is 1,675,133,810 bytes (sha256 `3eae03c2…0933d1f36a`, recorded in `data/SOURCES.md`) and contains 2,260,668 loans, which matches the published count. The raw file is never modified; `lcrisk ingest` derives everything from it.

3.2 Population and target

Step	Loans in	Excluded	Loans out
Valid loan id and issue date (drops summary lines)	2,260,701	33	2,260,668
36-month term	2,260,668	650,914	1,609,754
Issued on or before December 2015	1,609,754	988,585	621,169
Individual application	621,169	239	620,930
Final outcome observed (paid or charged off)	620,930	147	620,783

The issue-date cut makes every loan old enough to have run its full term before the data snapshot, so every outcome is final. Without it, recent vintages would be biased: their early defaults would be visible, but their full repayments would not yet be. Only 147 loans were still current or late at the snapshot, so the cut is clean.

Default (bad = 1): loan status *Charged Off*, *Default*, or the legacy *Does not meet the credit policy. Status: Charged Off*. **Good (bad = 0):** *Fully Paid* and its legacy equivalent. The population default rate is 13.94%.

3.3 Features

Features come from an explicit allowlist of fields known at application (`ingest.py`). Everything recorded after origination is excluded: payments, recoveries, the last FICO pull, hardship and settlement flags. Excluding by allowlist, not denylist, means a new post-origination field added to the source cannot leak in by accident. Three further groups are excluded on purpose:

- **Lending Club's grade, sub-grade and interest rate.** These are the output of Lending Club's own model. Using them would make ours a copy of theirs; instead they serve as a benchmark (Section 6.4). For the same reason the instalment amount, which is computed from the interest rate, is not used.
- **Geography** (state, three-digit zip). Fair-lending law treats location as a potential proxy for protected characteristics; it is left out rather than tested and defended.
- **Free text** (job title, loan description). It would need its own review.

Five features are derived: credit history length at application, employment length in years, the FICO band midpoint, and two affordability ratios, loan amount and revolving balance to annual income.

3.4 Data availability and the development window

Lending Club added sixteen bureau fields (total credit limits, bankcard utilisation and headroom, mortgage accounts, recent account openings) part-way through 2012. They are absent before 2012Q3, 43% populated that quarter, and complete from 2012Q4. Developing on 2007–2013 would have meant either dropping these fields or letting their missingness stand in for the issue date. The development window is therefore **2012Q4–2013Q4**, the period a lender would actually redevelop on. The earlier vintages are used only to calibrate the stress scenario (Section 8).

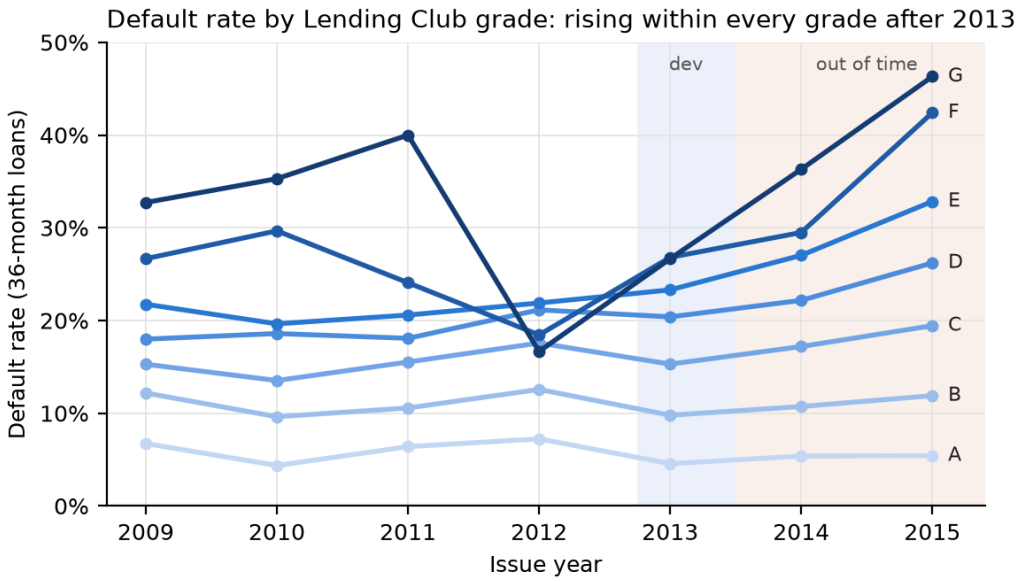


Figure 1. Observed default rate by Lending Club grade and issue year. From 2013 the default rate rises inside every grade, most steeply in the riskiest.

4. Methodology

4.1 Samples

The development window is split at random, stratified on the target, into training (70%, 80,922 loans) and test (30%, 34,681 loans). The 2014 and 2015 vintages are out-of-time samples scored separately. Binning, feature selection, master-scale boundaries and challenger hyperparameters were all fixed before either out-of-time sample was scored.

4.2 Champion: weight-of-evidence logistic scorecard

Binning. Each numeric feature starts from up to ten quantile bins. Adjacent bins are merged until (a) every bin holds at least 5% of the training sample and (b) the default rate moves monotonically across bins. Monotonicity is imposed because a scorecard that rewards a higher debt-to-income ratio over some range is hard to defend to a credit committee, and the reversal is usually noise. Missing values form their own bin

when they cover at least 0.5% of the sample, and otherwise score neutral (WoE = 0) rather than inherit a weight set by a handful of loans. Categories under 2% are pooled. WoE is ln(share of goods / share of bads), so positive WoE means lower risk.

Selection. Features need an information value of at least 0.02. Going down the IV ranking, a feature is dropped if its WoE correlates above 0.6 with one already kept. That leaves `avg_cur_bal`, `total_bc_limit` and `tot_cur_bal`, among others, behind `tot_hi_cred_lim` and `bc_open_to_buy`. A logistic regression is then fitted, and any feature with a wrong-signed (positive) coefficient or p > 0.01 is removed one at a time until none remain. None were removed at this stage.

Scaling. Scores are set to 600 points at good:bad odds of 50:1, with 20 points to double the odds. The intercept is shared equally across features, so each bin carries a whole number of points and the points add up to the score (Appendix A).

Feature	Coefficient	Std. error	IV	VIF
Intercept	−1.959	0.011		
FICO (band midpoint)	−0.539	0.040	0.124	1.57
Annual income	−0.409	0.046	0.103	1.84
Total high credit limit	−0.588	0.053	0.078	1.68
Bankcard open-to-buy	−0.356	0.057	0.075	1.92
Debt-to-income	−0.531	0.054	0.049	1.23
Loan to income	−0.739	0.057	0.043	1.24
Accounts opened, past 24 months	−0.971	0.058	0.042	1.19
% bankcards above 75% utilised	−0.330	0.072	0.041	1.78
Months since most recent enquiry	−0.943	0.062	0.033	1.10
Age of oldest revolving account	−0.281	0.064	0.032	1.15

All coefficients are negative, as required under this WoE convention, and all have |z| > 4. Every feature is a familiar credit-risk driver with an intuitive direction (Figure 7): high FICO, income, credit limits and headroom lower risk; leverage, recent credit-seeking and thin or young files raise it. Two missing-value bins are economically meaningful and kept: no bankcard at all (`bc_open_to_buy` missing, 15.3% default rate) is high risk, and no enquiry on file (`mths_since_recent_inq` missing, 9.4%) is the lowest-risk group of that feature.

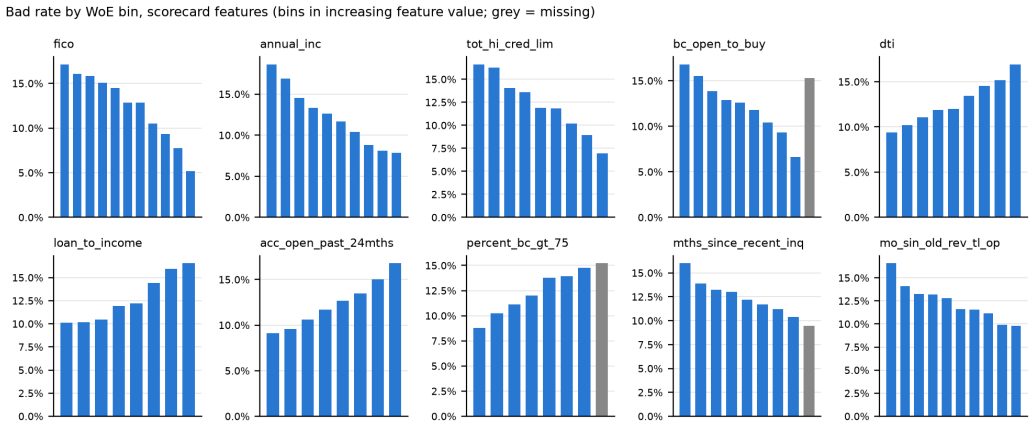


Figure 7. Training default rate by bin for each scorecard feature, bins in increasing feature value; grey bars are missing-value bins.

Master scale. PDs map to eight rating grades with fixed lifetime-PD boundaries: R1 < 4%, R2 4–6%, R3 6–8%, R4 8–11%, R5 11–15%, R6 15–20%, R7 20–27%, R8 ≥ 27%.

4.3 Challenger: LightGBM

The challenger uses all 40 allowlisted features in raw form, with categorical handling for home ownership, verification status and purpose. Seventeen features carry monotone constraints where the direction is not in doubt (FICO, debt-to-income, enquiries, utilisation, income, delinquencies and others; `challenger.MONOTONE`); a check confirms each constraint holds for every one of 2,000 sampled 2015 loans. Hyperparameters were chosen on a time-based hold-out inside the development window (train to 2013Q3, validate 2013Q4) from a seven-point grid. The grid was flat (hold-out Gini 0.350–0.354), so the choice is not fragile. The best setting, 15 leaves per tree and at least 1,000 loans per leaf, is also the most conservative; early stopping ended at 389 trees.

5. Performance on the development window

	Gini train	Gini test	KS test	Brier test	Hosmer–Lemeshow p (test)
Scorecard	0.315	0.310	0.226	0.1049	0.65
LightGBM	0.431	0.352	0.251	0.1037	0.19

The scorecard's train–test gap is half a Gini point. LightGBM's is eight: it fits the training sample closely. The test and out-of-time results below are the relevant measures, and they show no decay, so the gap is not a generalisation problem. It is still a reason to prefer the more conservative of the tuned settings when refitting. On the test sample, no master-scale grade is rejected by the binomial or Jeffreys test for either model.

6. Validation results

6.1 Discrimination over time

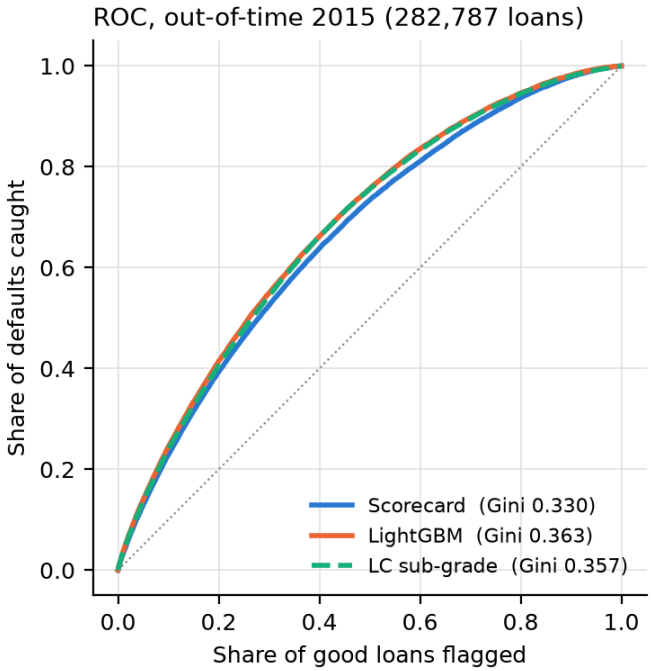


Figure 2. ROC curves on the 2015 out-of-time sample.

Neither model's ranking power decays out of time; it slightly improves (Figure 6), probably because the later vintages include more loans in the riskier grades, where separation is easier. Every out-of-time quarter's scorecard Gini sits within the development range (0.289–0.326) or above it. LightGBM's dips once, to 0.338 in 2014Q1, just below its development low of 0.342.

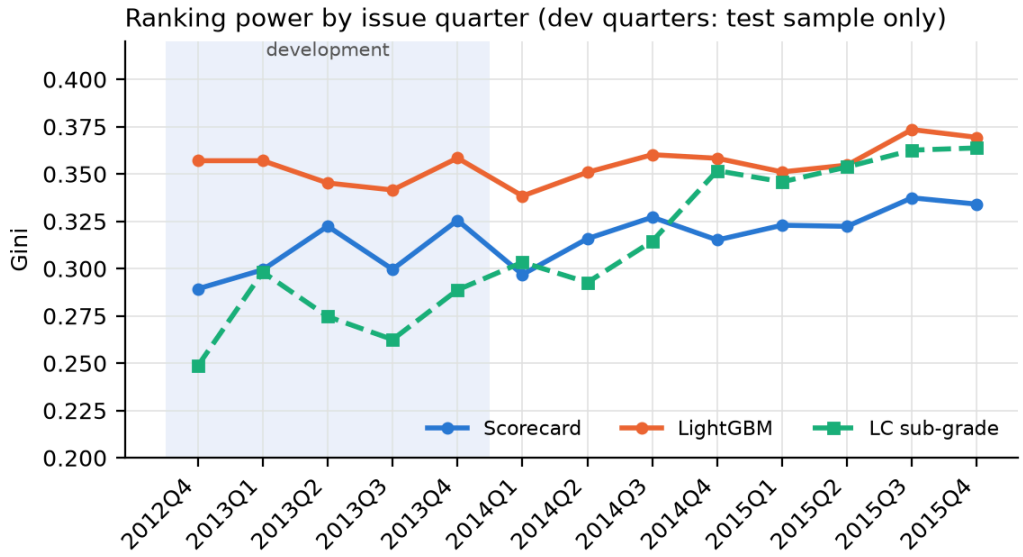


Figure 6. Gini by issue quarter. Development quarters show the test sample only.

6.2 Calibration

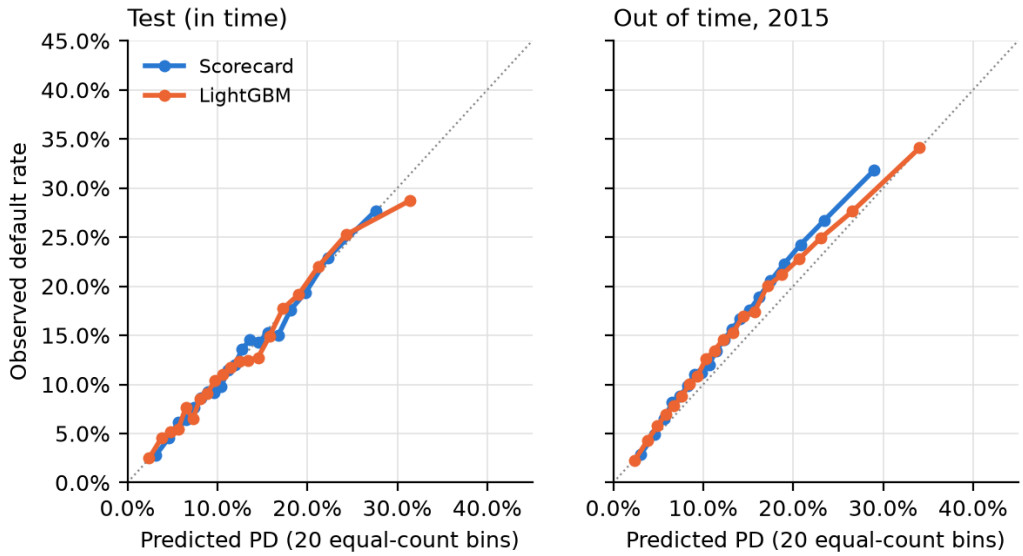


Figure 4. Predicted PD against observed default rate in 20 equal-count bins.

In time, both models sit on the diagonal. In 2015, both sit above it across the whole range: the shape of the curve is right but its level is wrong.

Sample	Mean PD (scorecard)	Observed	HL p (scorecard)	Grades rejected, Jeffreys 1%
Test	12.4%	12.4%	0.65	0 of 8
Out of time 2014	13.0%	13.7%	< 0.001	5 of 8
Out of time 2015	12.9%	14.9%	< 0.001	7 of 8

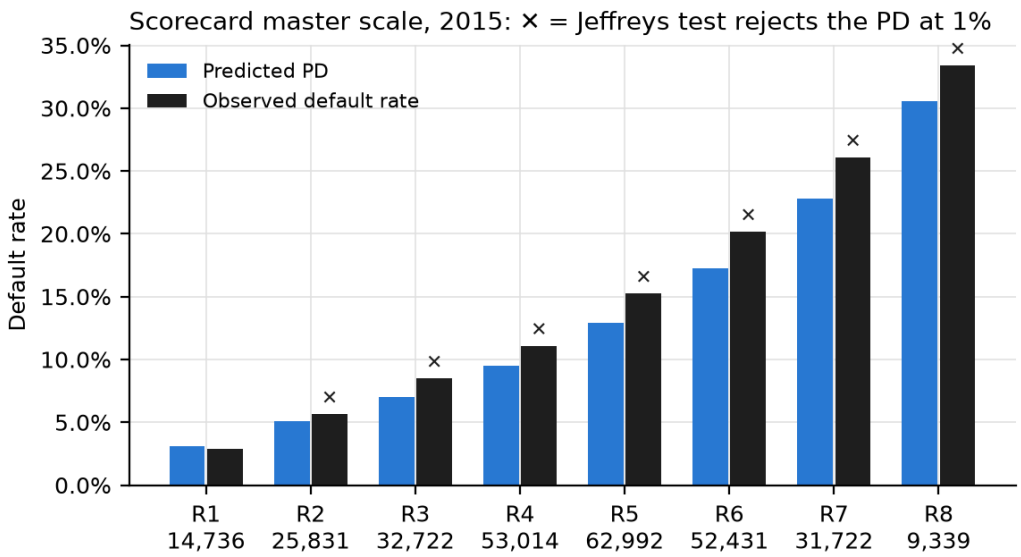


Figure 5. Scorecard master scale on 2015 loans. Only R1 passes.

6.3 Stability, and what it missed

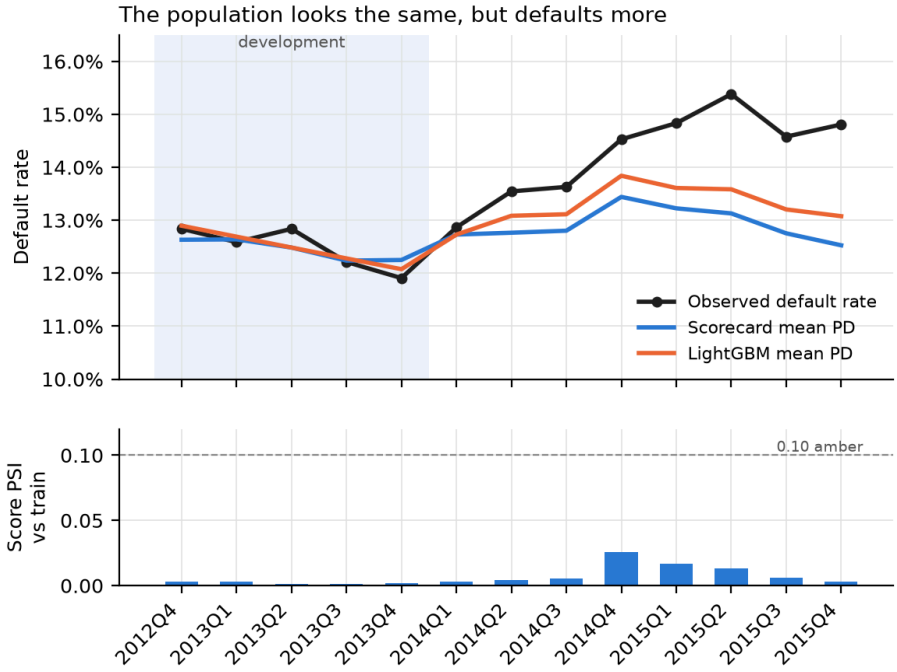


Figure 3. Mean predicted PD against observed default rate by issue quarter (top), and the score's population stability index against the training sample (bottom).

The score PSI peaks at 0.026 (2014Q4), far below the 0.10 amber threshold. Feature-level stability (CSI) is green for every feature except verification status (0.17 in 2014, 0.23 in 2015, amber), which reflects a change in Lending Club's income-verification practice. That feature is not in the scorecard (IV 0.003), and it contributes 1.8% of LightGBM's explanatory weight.

This is the central validation finding. The distribution of applicants barely moved, but the default rate given those characteristics did. That kind of drift is invisible to PSI by construction. Plausible causes are a looser macro-credit environment and adverse selection as competition for prime borrowers grew. Neither is observable in the application data. Monitoring that relies on PSI would have reported a healthy model throughout 2014–2015.

6.4 Benchmarking against Lending Club's own grade

Lending Club's sub-grade is a natural benchmark: it is the lender's own risk ranking, built with information this model does not have (the full bureau report, internal fraud and verification checks). On the development window it ranks worse than the scorecard (Gini 0.277 vs 0.310). It improves steadily after that and by 2015 matches LightGBM (0.357). Lending Club rebuilt its model on growing data during this period. A model frozen at 2013 falls behind one that keeps learning, which is a further argument for a defined redevelopment cycle.

6.5 Recalibration test

An intercept-only recalibration is the standard remedy when ranking holds and level drifts. To test it fairly, the shift is fitted on the most recent fully observed year (2014) and applied to the next (2015), as a bank would at an annual review.

Model	Mean PD before → after	Observed 2015	HL statistic before → after	Grades rejected after
Scorecard (+0.068 log-odds)	12.9% → 13.6%	14.9%	1,114 → 448	6 of 8
LightGBM (+0.043 log-odds)	13.3% → 13.8%	14.9%	763 → 423	6 of 8

The recalibration helps but lags: 2015 was worse than 2014. A one-year-old recalibration is not enough when conditions are moving, so the monitoring plan (Section 9) uses triggers.

6.6 Champion versus challenger

	Scorecard	LightGBM
Gini, out of time 2015	0.330	0.363
Bad rate of approved loans at 20% decline rate (2015)	12.04%	11.76%
Defaults avoided at 20% decline rate (2015)	14,852	15,480 (+628)
Features	10	40
Explanation	Points table; exact	TreeSHAP; per loan
Same top decline reason as scorecard		29% (76% within its top three)

At the average realised loss per 2015 default (about $6,500), LightGBM's extra 628 avoided defaults are worth about $4.1 million, 0.11% of the $3.6 billion lent. That is material. Against it, LightGBM's top decline reason is a feature the scorecard does not use 15% of the time, which would need a new adverse-action reason framework. It also spreads weight across many correlated bureau fields (Appendix B), so any single explanation is less stable. The recommendation is to keep the scorecard as champion and run LightGBM alongside it as a challenger and early-warning benchmark. If the gap persists over two further vintages and a reason-code framework is approved, LightGBM should be promoted.

7. Risk parameters and expected loss

Expected loss is PD × LGD × EAD per loan. LGD, EAD and default timing are estimated per master-scale grade on development-window defaults, whose recovery workouts were complete by the data snapshot.

- **EAD at default** = amount funded – principal repaid. It is 50% of the amount funded in R1 rising to 64% in R8, because riskier loans default earlier in their amortisation.
- **LGD** = 1 – (recoveries – collection fees) / EAD, exposure-weighted: 0.87–0.90 by grade. That is typical for unsecured consumer debt sold to collection buyers. Defaults that ended in a debt settlement (11%)

lose less, about 0.70.

- **12-month share.** Of lifetime defaults, 24% (R1) to 41% (R8) miss their first payment within 12 months of issue. This gives an empirical one-year PD for the capital calculation.

Lifetime EL backtest	Predicted EL	Realised loss	Predicted ÷ realised
2014, scorecard	$126.8m (6.20%)	$136.9m (6.69%)	0.93
2014, scorecard recalibrated	$135.1m (6.60%)		0.99
2015, scorecard	$223.0m (6.16%)	$273.8m (7.56%)	0.81
2015, scorecard recalibrated	$237.5m (6.56%)		0.87

The EL shortfall in 2015 follows from the PD shortfall: the LGD and EAD parameters themselves held up. One caution runs the other way: recoveries on late 2015 defaults may not have been fully booked at the snapshot, which would overstate realised loss slightly.

8. Stress testing and capital

Downturn calibration. Each 2007–2009 loan is given its sub-grade's development-window default rate as a PD. Those PDs average 14.2%; the crisis vintages actually defaulted at 16.5% (8,235 loans). Holding the sub-grade fixed removes the change in borrower mix and leaves the downturn itself, a shift of **+0.19 on the log-odds scale**. Crisis-vintage LGD was 5.4% higher than in the development window, which gives the downturn LGD factor. This is an approximate severity: Lending Club re-calibrated its grades over the period.

Scenario, 2015 book ($3.62bn funded)	Mean PD	Mean LGD	Lifetime EL	EL rate
Baseline (recalibrated PD)	13.6%	0.885	$237.5m	6.56%
Downturn PD	15.9%	0.886	$282.3m	7.80%
Downturn PD and LGD	15.9%	0.934	$297.4m	8.21%
Realised	*14.9%*		*$273.8m*	*7.56%*

The realised 2015 loss rate lies between the baseline and the downturn scenarios. In loss terms, the 2015 vintage experienced most of a crisis-calibrated stress, in a year without a recession. That is the clearest sign of how far underwriting conditions moved.

Capital. Basel's single-factor (Vasicek) formula for other retail exposures is applied to the one-year PD, with the regulatory asset correlation (3–16%) and downturn LGD. Unexpected loss at 99.9% is $373.5m, **10.3% of the amount funded**, on top of one-year expected loss of $135.3m. At 99% it is 6.4%. These are illustrative: there is no maturity adjustment, and EAD is taken at origination.

9. Monitoring plan

Metric	Frequency	Amber	Red	Action on red
Score PSI vs development	Monthly	≥ 0.10	≥ 0.25	Investigate; review cut-off
Feature CSI	Monthly	≥ 0.10	≥ 0.25	Investigate source change
30+ days past due at 6 months on book, predicted vs observed	Monthly, by vintage	Ratio > 1.10	> 1.20	Recalibrate intercept
Mean PD vs observed default rate, matured cohorts	Quarterly	Jeffreys rejects ≥ 2 grades	≥ 4 grades	Recalibrate; escalate
Gini on matured cohorts	Quarterly	< 0.28	< 0.25	Redevelopment review
Challenger Gini gap	Quarterly	> 0.04 for 2 quarters		Champion/challenger review

The early-delinquency backtest is the key addition. The full 36-month outcome for a vintage is known only three years after the loans are made. Section 6.3 shows that the distributional checks would not have flagged the problem in the meantime.

10. Findings log

ID	Severity	Finding	Required action
F1	High	PD under-predicts out of time: 2015 mean PD 12.9% vs 14.9% observed; Jeffreys rejects 7 of 8 grades	Recalibrate before use; adopt the trigger-based recalibration in Section 9
F2	High	Population stability metrics stayed green throughout the drift (score PSI ≤ 0.026)	Add the early-delinquency backtest to monitoring; do not rely on PSI alone
F3	Medium	Annual recalibration lags a deteriorating environment (6 of 8 grades still rejected after the 2014-based fix)	Recalibrate on triggers, not a calendar
F4	Medium	Development data are approved loans only; no reject inference	Treat PDs for applicant profiles outside the approved range as unvalidated; consider reject inference at redevelopment
F5	Medium	Challenger outperforms by about 4 Gini points but its decline reasons differ from the champion's	Build and approve a SHAP-based reason-code framework before any promotion
F6	Low	LightGBM in-sample Gini 0.43 vs 0.35 on test	Prefer stronger regularisation at refit; no decay seen out of time
F7	Low	Verification status CSI amber (0.23 in 2015)	Not in the champion; monitor its use in the challenger
F8	Low	Downturn severity is calibrated through Lending Club sub-grades, which were redefined over time	Cross-check against an external macro scenario at the next review
F9	Low	Recoveries on late 2015 defaults may be incomplete at the data snapshot	Re-estimate LGD on a later snapshot if one becomes available

11. Limitations and assumptions

- **Public data, single lender.** Lending Club's applicants, policy and pricing are not a bank's. The method transfers; the parameters do not.
- **Lifetime target.** The PD is a 36-month lifetime PD. The 12-month PD used for capital comes from the observed timing of defaults by grade, assuming that timing is stable.
- **Selection.** Only approved and funded loans are observed. The model has not been tested on applicants Lending Club declined.
- **No macro variables.** The model is through-the-cycle by construction; the stress scenario supplies the cycle.

Appendix A. Scorecard points

The scorecard is 10 features and 92 bins. Score = sum of the points for each feature's bin; 600 = 50:1 good:bad odds; +20 points doubles the odds. The full table is in `reports/tables/scorecard_points.csv`. Selected rows:

Feature	Bin	Training loans	Default rate	Points
FICO	≤ 662	5,789	17.1%	48
FICO	697–707	9,460	10.5%	57
FICO	> 737	6,974	5.2%	69
Accounts opened, past 24 months	0	4,692	9.1%	64
Accounts opened, past 24 months	> 7	7,313	16.8%	44
Months since most recent enquiry	≤ 1	13,459	16.0%	46
Months since most recent enquiry	none on file	9,209	9.4%	63
Loan to income	≤ 0.109	16,216	10.1%	59
Loan to income	> 0.341	8,058	16.6%	47

Appendix B. Feature importance, both models

Share of mean absolute contribution to the log-odds on 20,000 loans from 2015. The full table is in `reports/tables/importance.csv`.

Feature	Scorecard	LightGBM
FICO	14.9%	10.5%
Annual income	9.8%	8.8%
Accounts opened, past 24 months	14.8%	6.8%
Loan to income	11.3%	5.9%
Debt-to-income	9.2%	5.8%
Accounts opened, past 12 months	—	4.9%
Loan purpose	—	4.8%
Months since most recent enquiry	12.1%	4.7%
Total high credit limit	11.5%	3.1%
31 other features (LightGBM) / 3 other features (scorecard)	16.4%	44.7%